

Complete Publication List

Oliver Gutsche

May 08, 2026

1. Search for a new heavy scalar resonance decaying into the Higgs boson and a new scalar particle in the $b\bar{b}b\bar{b}$ final state using proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.02848](#) | 2026-05-04

2. Observation of the rare decay $\eta \rightarrow \mu^+ \mu^- e^+ e^-$

Hayrapetyan, Aram et al.

[arXiv:2605.00615](#) | 2026-05-01

3. Search for light charged Higgs bosons decaying to charm and strange quarks in $t\bar{t}$ events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.27795](#) | 2026-05-01

4. Improved results on Higgs boson pair production in the $4b$ final state

Hayrapetyan, Aram et al.

[arXiv:2604.27044](#) | 2026-04-29

5. Search for electroweakinos in compressed-spectrum scenarios with low-momentum isolated tracks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.25604](#) | 2026-04-28

6. Simultaneous measurements of N-subjettiness observables in jets from gluons and light-flavour quarks, and in decays of boosted W bosons and top quarks

Hayrapetyan, Aram et al.

[arXiv:2604.25538](#) | 2026-04-28

7. Measurement of the $Z \rightarrow \mu^+ \mu^-$ angular coefficients in pp collisions at $\sqrt{s} = 13$ TeV as functions of transverse momentum and rapidity

Gevorgyan, Arzunik et al.

[arXiv:2604.25678](#) | 2026-04-28

8. Search for associated production of a Higgs boson and two vector bosons via vector boson scattering at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.24531](#) | 2026-04-27

9. Search for dark matter produced in association with a dark Higgs boson decaying into a bottom quark-antiquark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.20999](#) | 2026-04-22

10. Search for the single production of vector-like quarks decaying into a W boson and a b quark using single-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.17564](#) | 2026-04-19

11. Highly boosted dielectron identification in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.13320](#) | 2026-04-14

12. Search for heavy resonances decaying into four-lepton final states via light bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.14236](#) | 2026-04-14

13. A search for microscopic black holes, string balls, and sphalerons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.10732](#) | 2026-04-12

14. Search for a new heavy resonance decaying to a top quark and a neutral scalar boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.10729](#) | 2026-04-12

15. Particle transformers for identifying Lorentz-boosted Higgs bosons decaying to a pair of W bosons

Hayrapetyan, Aram et al.

[arXiv:2604.09809](#) | 2026-04-10

16. Observation of nuclear suppression in coherent $\Upsilon(1S)$ photoproduction off heavy nuclei at the LHC

Belyaev, Andrey et al.

[arXiv:2604.05814](#) | 2026-04-07

17. Search for soft unclustered energy patterns produced in association with a W or Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.05996](#) | 2026-04-07

18. Evidence of $ZZ\gamma$ production and observation of $4\ell\gamma$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.02594](#) | 2026-04-03

19. Search for Higgs boson pair production in the $b\bar{b}WW$ decay channel with two leptons in the final state using proton-proton collision data at $\sqrt{s} = 13.6$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2604.02127](#) | 2026-04-02

20. Measurement of dijet angular distributions and search for beyond the standard model physics in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2603.25458](#) | 2026-03-26

21. Search for new particles decaying into top quark-antiquark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2603.23454](#) | 2026-03-24

22. Search for low-mass vector and scalar resonances decaying into a quark-antiquark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.21965](#) | 2026-03-23

23. Search for Higgs boson production at high transverse momentum in the WW decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2603.22233](#) | 2026-03-23

24. Measurement of the jet mass in hadronic decays of boosted W bosons at 13 TeV and extraction of the W boson mass

Hayrapetyan, Aram et al.
[arXiv:2603.19963](#) | 2026-03-20

25. Jet peak shapes based on two-particle angular correlations in lead-lead collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Belyaev, Andrey et al.
[arXiv:2603.14385](#) | 2026-03-15

26. Measurement of the t-channel single top quark cross section in proton-proton collisions at $\sqrt{s} = 5.02$ TeV

Hayrapetyan, Aram et al.
[arXiv:2603.13592](#) | 2026-03-13

27. Search for Z' bosons decaying into charginos in final states with two oppositely charged leptons and missing transverse momentum in pp collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2603.11035](#) | 2026-03-11

28. Measurement of angular correlations inside jets induced by gluon polarization in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.
[arXiv:2603.03689](#) | 2026-03-04

29. Search for a narrow resonance with a mass between 10 and 70 GeV decaying to a pair of photons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2603.03250](#) | 2026-03-03

30. Combination of ATLAS and CMS searches for Higgs boson pair production at $\sqrt{s} = 13$ TeV

Aad, Georges et al.
[arXiv:2602.23991](#) | 2026-02-27

31. Search for pair production of heavy resonances in final states with a photon and large-radius jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2602.20477](#) | 2026-02-24

32. System-size dependence of charged-particle suppression in ultrarelativistic nucleus-nucleus collisions

Belyaev, Andrey et al.
[arXiv:2602.21325](#) | 2026-02-24

33. Observation of the jet diffusion wake using dijets in heavy ion collisions

Hayrapetyan, Aram et al.
[arXiv:2602.19431](#) | 2026-02-23

34. First measurement of jet axis decorrelation with photon-tagged jets in pp and PbPb collisions at 5.02 TeV

Hayrapetyan, Aram et al.
[arXiv:2602.18279](#) | 2026-02-20

35. Search for a new resonance decaying to a Higgs boson and a scalar boson in events with two b jets and two Z bosons in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18223](#) | 2026-02-20

36. Combined measurements and interpretations of Higgs boson production and decay in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18611](#) | 2026-02-20

37. Measurement of event shape variables using charged particles inside jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.17509](#) | 2026-02-19

38. Building an AI-native Research Ecosystem for Experimental Particle Physics: A Community Vision

Aarrestad, Thea Klæboe et al.

[arXiv:2602.17582](#) | 2026-02-19

39. Interpretable Models for Workflow Differentiation in High-Performance Scientific Networks

Giannakou, Anna et al.

(2026) 349 | DOI:10.1109/ICNC68183.2026.11416957 | 2026

40. Influence of strangeness on the anisotropic flow of prompt $D^{\pm}_s$ mesons in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Belyaev, Andrey et al.

[arXiv:2602.14221](#) | 2026-02-15

41. High-level hadronic tau lepton triggers of the CMS experiment in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

JINST 21 (2026) P04002 | [arXiv:2602.11359](#) | DOI:10.1088/1748-0221/21/04/P04002 | 2026-02-11

42. Measurement of B meson production fraction ratios in proton-proton collisions at $\sqrt{s} = 13$ TeV using open-charm and charmonium decays

Hayrapetyan, Aram et al.

[arXiv:2602.10270](#) | 2026-02-10

43. Probing early parton emissions in heavy ion collisions using the Lund jet plane

Belyaev, Andrey et al.

[arXiv:2602.09271](#) | 2026-02-09

44. Observation of a family of all-charm tetraquarks

Hayrapetyan, Aram et al.

[arXiv:2602.02252](#) | 2026-02-02

45. Observation of $\Upsilon(1S) + Z$ associated production and measurement of the effective double-parton scattering cross section in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.02770](#) | 2026-02-02

46. Search for heavy scalar resonances decaying to Lorentz-boosted Higgs and Higgs-like bosons in the $b\bar{b}4q$ final state at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.00273](#) | 2026-01-30

47. Measurement and effective field theory interpretation of the photon-fusion production cross section of a pair of W bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.21574](#) | 2026-01-29

48. Search for heavy long-lived charged particles with level-1 trigger scouting data from proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.20063](#) | 2026-01-27

49. Measurement of the $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ differential cross sections in pp collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.20023](#) | 2026-01-27

50. Search for heavy resonances decaying into two Higgs bosons in the $b\bar{b}\tau^+\tau^-$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.20011](#) | 2026-01-27

51. Evaluating Application Characteristics for GPU Portability Layer Selection

Atif, Mohammad et al.

[arXiv:2601.17526](#) | 2026-01-24

52. Search for the pair production of long-lived supersymmetric partners of the tau lepton in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.17576](#) | 2026-01-24

53. Full event interpretation with machine-learning-based particle-flow reconstruction in the CMS detector

Hayrapetyan, Aram et al.

[arXiv:2601.17554](#) | 2026-01-24

54. Strategy and performance of the CMS long-lived particle trigger program in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.17544](#) | 2026-01-24

55. Measurement of the $Z\gamma$ production cross section and search for anomalous neutral triple gauge couplings in pp collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.14102](#) | 2026-01-20

56. Combination of searches for heavy vector boson resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.12583](#) | 2026-01-18

57. Search for dark matter produced in association with a Higgs boson decaying to bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.11330](#) | 2026-01-16

58. Measurement of the Higgs boson total decay width using the $H \rightarrow WW \rightarrow e\nu\mu\nu$ decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.05168](#) | 2026-01-08

59. Search for a boosted Higgs boson decaying to bottom quark pairs in association with a W or Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.05362](#) | 2026-01-08

60. Search for exotic Higgs boson decays $H \rightarrow \mathcal{A}\mathcal{A}$ with $\mathcal{A}\mathcal{A} \rightarrow \gamma\gamma$ in events with a semi-merged topology in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.00183](#) | 2026-01-01

61. Search for charged Higgs bosons decaying into top and bottom quarks in lepton+jets final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2512.24471](#) | 2025-12-30

62. Machine-learning techniques for model-independent searches in dijet final states

Hayrapetyan, Aram et al.

[arXiv:2512.20395](#) | [DOI:10.5281/zenodo.16656501](#) | 2025-12-23

63. Characterization of the quantum state of top quark pairs produced in proton-proton collisions at $\sqrt{s}=13\text{TeV}$ using the beam and helicity bases

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) L051101](#) | [arXiv:2512.17557](#) | [DOI:10.1103/hkgw-dztk](#) | 2025-12-19

64. Measurement of $Z\gamma$ production in proton-proton collisions at $\sqrt{s} = 13.6$ TeV and constraints on neutral triple gauge couplings

Hayrapetyan, Aram et al.

[arXiv:2512.08582](#) | 2025-12-09

65. Measurements of electroweak production of a photon in association with two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2512.00502](#) | 2025-11-29

66. Search for light pseudoscalar bosons, pair-produced in Higgs boson decays in the four-electron final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2511.19563](#) | 2025-11-24

67. Search for new physics in the final state with a single photon and large missing transverse momentum in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 072020](#) | [arXiv:2511.17310](#) | [DOI:10.1103/f9r8-styj](#) | 2025-11-21

68. Search for Higgsinos in final states with low-momentum lepton-track pairs at 13 TeV

Hayrapetyan, Aram et al.

[arXiv:2511.16394](#) | 2025-11-20

69. Search for heavy $H\gamma$ and $Z\gamma$ resonances with a bottom quark-antiquark pair in the final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2511.14583](#) | 2025-11-18

70. Search for low-mass hidden-valley dark showers with non-prompt muon pairs in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 03 \(2026\) 189](#) | [arXiv:2511.11888](#) | [DOI:10.1007/JHEP03\(2026\)189](#) | 2025-11-14

71. Search for a new neutral gauge boson produced in association with one or two b jets and decaying into a pair of muons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Chekhovsky, Vladimir et al.

[arXiv:2511.11853](#) | 2025-11-14

72. Search for long-lived particles using displaced vertices with low-momentum tracks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2511.08212](#) | 2025-11-11

73. Identification of tau leptons using a convolutional neural network with domain adaptation

Hayrapetyan, Aram et al.

[JINST 20 \(2025\) P12032](#) | [arXiv:2511.05468](#) | [DOI:10.1088/1748-0221/20/12/P12032](#) | 2025-11-07

74. Jet fragmentation function and groomed substructure of bottom quark jets in proton-proton collisions at 5.02 TeV

Chekhovsky, Vladimir et al.

[JHEP 04 \(2026\) 147](#) | [arXiv:2511.10666](#) | [DOI:10.1007/JHEP04\(2026\)147](#) | 2025-11-06

75. Inclusive and differential measurements of the $t\bar{t}\gamma$ cross section and the $t\bar{t}\gamma / t\bar{t}$ cross section ratio in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 04 \(2026\) 086](#) | [arXiv:2511.01995](#) | [DOI:10.1007/JHEP04\(2026\)086](#) | 2025-11-03

76. Characterizing the initial state and dynamical evolution in XeXe and PbPb collisions using multiparticle cumulants

Hayrapetyan, Aram et al.

[Phys.Lett.B 876 \(2026\) 140359](#) | [arXiv:2510.26766](#) | [DOI:10.1016/j.physletb.2026.140359](#) | 2025-10-30

77. Search for single production of a vector-like T quark decaying to a top quark and a neutral scalar boson in the lepton+jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2510.25874](#) | 2025-10-29

78. Measurement of the ratio of the $B_c^+ \rightarrow J/\psi \tau^+ \nu_\tau$ and $B_c^+ \rightarrow J/\psi \mu^+ \nu_\mu$ branching fractions using three-prong τ lepton decays

Chekhovsky, Vladimir et al.

[arXiv:2510.21559](#) | 2025-10-24

79. Search for dijet resonances with data scouting in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 04 \(2026\) 133](#) | [arXiv:2510.21641](#) | [DOI:10.1007/JHEP04\(2026\)133](#) | 2025-10-24

80. Observation of tWZ Production at the CMS Experiment

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 081802](#) | [arXiv:2510.19080](#) | [DOI:10.1103/rk6w-1pcl](#) | 2025-10-21

81. Search for dark matter production in association with bottom quarks and a lepton pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Chekhovsky, Vladimir et al.

[arXiv:2510.12396](#) | 2025-10-14

82. Performance of heavy-flavour jet identification in Lorentz-boosted topologies in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JINST 20 \(2025\) P11006](#) | [arXiv:2510.10228](#) | [DOI:10.1088/1748-0221/20/11/P11006](#) | 2025-10-11

83. Observation of Suppressed Charged-Particle Production in Ultrarelativistic Oxygen-Oxygen Collisions

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 162301](#) | [arXiv:2510.09864](#) | [DOI:10.1103/89sf-9t1x](#) | 2025-10-10

84. Combination of searches for nonresonant Higgs boson pair production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2510.07527](#) | 2025-10-08

85. Wasserstein normalized autoencoder for anomaly detection

Hayrapetyan, Aram et al.

[arXiv:2510.02168](#) | 2025-10-02

86. Observation of long-range collective flow in OO and NeNe collisions and implications for nuclear structure studies

Hayrapetyan, Aram et al.

[arXiv:2510.02580](#) | 2025-10-02

87. Vector boson scattering and anomalous quartic couplings in final states with $\ell\nu qq$ or $\ell\ell qq$ plus jets using proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 03 \(2026\) 022](#) | [arXiv:2510.00118](#) | [DOI:10.1007/JHEP03\(2026\)022](#) | 2025-09-30

88. Measurement of the dineutrino system kinematic variables in dileptonic top quark pair production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Chekhovsky, Vladimir et al.

[arXiv:2510.00160](#) | 2025-09-30

89. Simultaneous Probe of the Charm and Bottom Quark Yukawa Couplings Using $t\bar{t}H$ Events

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 011801](#) | [arXiv:2509.22535](#) | [DOI:10.1103/9nwb-splk](#) | 2025-09-26

90. Search for resonances decaying to an anomalous jet and a Higgs boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 86 \(2026\) 137](#) | [arXiv:2509.13635](#) | [DOI:10.1140/epjc/s10052-025-15176-9](#) | 2025-09-17

91. Measurements of $t\bar{t}W$ differential cross sections and the leptonic charge asymmetry at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 03 \(2026\) 083](#) | [arXiv:2509.13512](#) | [DOI:10.1007/JHEP03\(2026\)083](#) | 2025-09-16

92. Improving missing transverse momentum estimation with a deep neural network

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 072010](#) | [arXiv:2509.12012](#) | [DOI:10.1103/c4z7-tqvc](#) | 2025-09-15

93. Measurement of D^0 Meson Photoproduction in Ultraperipheral Heavy Ion Collisions

Chekhovsky, Vladimir et al.

[Phys.Rev.Lett. 136 \(2026\) 122303](#) | [arXiv:2509.08626](#) | [DOI:10.1103/lckg-sdh9](#) | 2025-09-10

94. Model-independent measurement of the Higgs boson associated production with two jets and decaying to a pair of W bosons in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 03 \(2026\) 073](#) | [arXiv:2509.07958](#) | [DOI:10.1007/JHEP03\(2026\)073](#) | 2025-09-09

95. Search for a Higgs boson produced in association with a charm quark and decaying to a W boson pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2508.14988](#) | 2025-08-20

96. General search for supersymmetric particles in scenarios with compressed mass spectra using proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[Phys.Rev.D 112 \(2025\) 112023](#) | [arXiv:2508.13900](#) | [DOI:10.1103/b26z-zmpy](#) | 2025-08-19

97. Search for a new scalar resonance decaying to a Higgs boson and another new scalar particle in the final state with two bottom quarks and two photons in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 12 \(2025\) 178](#) | [arXiv:2508.11494](#) | [DOI:10.1007/JHEP12\(2025\)178](#) | 2025-08-15

98. Search for charged lepton flavor violating Z and Z' boson decays in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 112 \(2025\) 112011](#) | [arXiv:2508.07512](#) | [DOI:10.1103/8lw4-6lyk](#) | 2025-08-11

99. Search for light pseudoscalar boson pairs produced from Higgs boson decays using the 4τ and $2\mu 2\tau$ final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 04 \(2026\) 132](#) | [arXiv:2508.06947](#) | [DOI:10.1007/JHEP04\(2026\)132](#) | 2025-08-09

100. Search for b-hadron decays to long-lived particles in the CMS endcap muon detectors

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 012009](#) | [arXiv:2508.06363](#) | [DOI:10.1103/7ldn-snzn](#) | 2025-08-08

101. First Exclusive Reconstruction of the B^{*+} , B^{*0} , and B_s^{*0} Mesons and Precise Measurement of Their Masses

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 031902](#) | [arXiv:2508.05820](#) | [DOI:10.1103/njf9-4zfv](#) | 2025-08-07

102. Probing the flavour structure of dimension-6 EFT operators in multilepton final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 03 \(2026\) 140](#) | [arXiv:2507.17498](#) | [DOI:10.1007/JHEP03\(2026\)140](#) | 2025-07-23

103. Search for resonant production of pairs of dijet resonances through broad mediators in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2507.17884](#) | 2025-07-23

104. Exploring small-angle emissions in charm quark jets in proton-proton collisions at $\sqrt{s} = 5.02$ TeV

Chekhovsky, Vladimir et al.

[arXiv:2507.13469](#) | 2025-07-17

105. Evidence of medium response to hard probes using correlations of Z bosons with hadrons in heavy ion collisions

Chekhovsky, Vladimir et al.

[Phys.Lett.B 874 \(2026\) 140120](#) | [arXiv:2507.09307](#) | [DOI:10.1016/j.physletb.2025.140120](#) | 2025-07-12

106. A method for correcting the substructure of multiprong jets using the Lund jet plane

Hayrapetyan, Aram et al.

[JHEP 11 \(2025\) 038](#) | [arXiv:2507.07775](#) | [DOI:10.1007/JHEP11\(2025\)038](#) | 2025-07-10

107. Search for heavy pseudoscalar and scalar bosons decaying to a top quark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 88 \(2025\) 127801](#) | [arXiv:2507.05119](#) | [DOI:10.1088/1361-6633/ae2207](#) | 2025-07-07

108. Search for the nonresonant and resonant production of a Higgs boson in association with an additional scalar boson in the $\gamma\gamma\tau\tau$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2506.23012](#) | 2025-06-28

109. Operation and performance of the CMS silicon strip tracker with proton-proton collisions at the CERN LHC

Hayrapetyan, Aram et al.

[JINST 20 \(2025\) P08027](#) | [arXiv:2506.17195](#) | [DOI:10.1088/1748-0221/20/08/P08027](#) | 2025-06-20

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167. Search for pair production of heavy particles decaying to a top quark and a gluon in the lepton+jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.D 111 \(2025\) 072003](#) | [arXiv:2409.17770](#) | [DOI:10.1103/PhysRevD.111.072003](#) | 2024-09-26

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[Phys.Lett.B 861 \(2025\) 139279](#) | [arXiv:2409.00838](#) | [DOI:10.1016/j.physletb.2025.139279](#) | 2024-09-01

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[JHEP 04 \(2025\) 162](#) | [arXiv:2408.03744](#) | [DOI:10.1007/JHEP04\(2025\)162](#) | 2024-08-07

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[Phys.Lett.B 861 \(2025\) 139210](#) | [arXiv:2407.13554](#) | [DOI:10.1016/j.physletb.2024.139210](#) | 2024-07-18

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[JHEP 02 \(2025\) 040](#) | [arXiv:2407.13872](#) | [DOI:10.1007/JHEP02\(2025\)040](#) | 2024-07-18

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Hayrapetyan, Aram et al.

[JHEP 02 \(2025\) 036](#) | [arXiv:2407.10717](#) | [DOI:10.1007/JHEP02\(2025\)036](#) | 2024-07-15

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Hayrapetyan, Aram et al.

[JHEP 02 \(2025\) 097](#) | [arXiv:2407.10896](#) | [DOI:10.1007/JHEP02\(2025\)097](#) | 2024-07-15

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Hayrapetyan, Aram et al.

[JHEP 12 \(2024\) 035](#) | [arXiv:2407.08012](#) | [DOI:10.1007/JHEP12\(2024\)035](#) | 2024-07-10

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[Phys.Rev.Lett. 134 \(2025\) 111903](#) | [arXiv:2407.05402](#) | [DOI:10.1103/PhysRevLett.134.111903](#) | 2024-07-07

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[Phys.Lett.B 858 \(2024\) 139044](#) | [arXiv:2406.14409](#) | [DOI:10.1016/j.physletb.2024.139044](#) | 2024-06-20

198. Search for a resonance decaying to a W boson and a photon in proton-proton collisions at $\sqrt{s} = 13$ TeV using leptonic W boson decays

Hayrapetyan, Aram et al.

199. Measurement of inclusive and differential cross sections for W^+W^- production in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

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[Phys.Lett.B 860 \(2025\) 139067](#) | [arXiv:2405.18149](#) | [DOI:10.1016/j.physletb.2024.139067](#) | 2024-05-28

203. Stairway to discovery: A report on the CMS programme of cross section measurements from millibarns to femtobarns

Hayrapetyan, Aram et al.

[Phys.Rept. 1115 \(2025\) 3](#) | [arXiv:2405.18661](#) | [DOI:10.1016/j.physrep.2024.11.005](#) | 2024-05-28

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[Phys.Rept. 1115 \(2025\) 2350](#) | [arXiv:2405.17605](#) | [DOI:10.1016/j.physrep.2024.09.012](#) | 2024-05-27

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[Phys.Lett.B 860 \(2025\) 139202](#) | [arXiv:2405.16566](#) | [DOI:10.1016/j.physletb.2024.139202](#) | 2024-05-26

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[Phys.Lett.B 857 \(2024\) 138979](#) | [arXiv:2405.14757](#) | [DOI:10.1016/j.physletb.2024.138979](#) | 2024-05-23

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[Phys.Rept. 1115 \(2025\) 2367](#) | [arXiv:2405.10785](#) | [DOI:10.1016/j.physrep.2024.11.007](#) | 2024-05-17

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[JHEP 08 \(2024\) 215](#) | [arXiv:2405.09320](#) | [DOI:10.1007/JHEP08\(2024\)215](#) | 2024-05-15

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[Phys.Rev.D 110 \(2024\) 072012](#) | [arXiv:2405.05071](#) | [DOI:10.1103/PhysRevD.110.072012](#) | 2024-05-08

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[Phys.Rev.Lett. 133 \(2024\) 161805](#) | [arXiv:2404.18298](#) | [DOI:10.1103/PhysRevLett.133.161805](#) | 2024-04-28

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[Phys.Rev.Lett. 133 \(2024\) 201803](#) | [arXiv:2404.02992](#) | [DOI:10.1103/PhysRevLett.133.201803](#) | 2024-04-03

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[Eur.Phys.J.C 84 \(2024\) 712](#) | [arXiv:2403.20241](#) | [DOI:10.1140/epjc/s10052-024-13021-z](#) | 2024-03-29

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[JHEP 07 \(2024\) 142](#) | [arXiv:2403.01556](#) | [DOI:10.1007/JHEP07\(2024\)142](#) | 2024-03-03

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[Phys.Rev.Lett. 132 \(2024\) 241802](#) | [arXiv:2402.18461](#) | [DOI:10.1103/PhysRevLett.132.241802](#) | 2024-02-28

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Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 012004](#) | [arXiv:2402.18658](#) | [DOI:10.1103/PhysRevD.110.012004](#) | 2024-02-28

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Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 012002](#) | [arXiv:2402.17738](#) | [DOI:10.1103/PhysRevD.110.012002](#) | 2024-02-27

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[Phys.Rev.D 109 \(2024\) 112005](#) | [arXiv:2402.15804](#) | [DOI:10.1103/PhysRevD.109.112005](#) | 2024-02-24

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240. Search for long-lived particles decaying to final states with a pair of muons in proton-proton collisions at $\sqrt{s} = 13.6 \text{ TeV}$

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[JHEP 05 \(2024\) 047](#) | [arXiv:2402.14491](#) | [DOI:10.1007/JHEP05\(2024\)047](#) | 2024-02-22

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[Phys.Rev.D 110 \(2024\) 052004](#) | [arXiv:2402.13808](#) | [DOI:10.1103/PhysRevD.110.052004](#) | 2024-02-21

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[Eur.Phys.J.C 84 \(2024\) 493](#) | [arXiv:2402.13358](#) | [DOI:10.1140/epjc/s10052-024-12727-4](#) | 2024-02-20

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[Phys.Rev.D 110 \(2024\) 012013](#) | [arXiv:2402.11098](#) | [DOI:10.1103/PhysRevD.110.012013](#) | 2024-02-16

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[Phys.Rev.Lett. 134 \(2025\) 131802](#) | [arXiv:2402.09932](#) | [DOI:10.1103/PhysRevLett.134.131802](#) | 2024-02-15

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[JHEP 02 \(2025\) 064](#) | [arXiv:2402.08486](#) | [DOI:10.1007/JHEP02\(2025\)064](#) | 2024-02-13

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[Phys.Rev.Lett. 132 \(2024\) 261902](#) | [arXiv:2402.08713](#) | [DOI:10.1103/PhysRevLett.132.261902](#) | 2024-02-13

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[Phys.Rev.D 109 \(2024\) 112003](#) | [arXiv:2402.08668](#) | [DOI:10.1103/PhysRevD.109.112003](#) | 2024-02-09

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Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112001](#) | [arXiv:2402.01888](#) | [DOI:10.1103/PhysRevD.109.112001](#) | 2024-02-02

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[Phys.Rev.D 110 \(2024\) 032007](#) | [arXiv:2402.01898](#) | [DOI:10.1103/PhysRevD.110.032007](#) | 2024-02-02

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[EPJ Web Conf. 295 \(2024\) 11003](#) | [arXiv:2401.14221](#) | [DOI:10.1051/epjconf/202429511003](#) | 2024-01-25

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[Phys.Rev.D 109 \(2024\) 112013](#) | [arXiv:2401.14494](#) | [DOI:10.1103/PhysRevD.109.112013](#) | 2024-01-25

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255. Test of lepton flavor universality in $B^{\pm} \rightarrow K^{\pm} \mu^+ \mu^-$ and $B^{\pm} \rightarrow K^{\pm} e^+ e^-$ decays in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

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[Rept.Prog.Phys. 87 \(2024\) 077801](#) | [arXiv:2401.06896](#) | [DOI:10.1088/1361-6633/ad4b9b](#) | 2024-01-12

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[Phys.Lett.B 855 \(2024\) 138815](#) | [arXiv:2312.11668](#) | [DOI:10.1016/j.physletb.2024.138815](#) | 2023-12-18

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[JHEP 2024 \(2024\) 106](#) | [arXiv:2311.11370](#) | [DOI:10.1007/JHEP02\(2024\)106](#) | 2023-11-19

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[Phys.Lett.B 850 \(2024\) 138478](#) | [arXiv:2311.03261](#) | [DOI:10.1016/j.physletb.2024.138478](#) | 2023-11-06

271. Search for high-mass exclusive diphoton production with tagged protons in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.D 110 \(2024\) 012010](#) | [arXiv:2311.02725](#) | [DOI:10.1103/PhysRevD.110.012010](#) | 2023-11-05

272. Search for an exotic decay of the Higgs boson into a Z boson and a pseudoscalar particle in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Lett.B 852 \(2024\) 138582](#) | [arXiv:2311.00130](#) | [DOI:10.1016/j.physletb.2024.138582](#) | 2023-10-31

273. Search for W' bosons decaying to a top and a bottom quark in leptonic final states in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 046](#) | [arXiv:2310.19893](#) | [DOI:10.1007/JHEP05\(2024\)046](#) | 2023-10-30

274. Search for Narrow Trijet Resonances in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 133 \(2024\) 011801](#) | [arXiv:2310.14023](#) | [DOI:10.1103/PhysRevLett.133.011801](#) | 2023-10-21

275. Search for central exclusive production of top quark pairs in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$ with tagged protons

Tumasyan, Armen et al.

[JHEP 06 \(2024\) 187](#) | [arXiv:2310.11231](#) | [DOI:10.1007/JHEP06\(2024\)187](#) | 2023-10-17

276. Observation of $WW\gamma$ production and search for $H\gamma$ production in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 121901](#) | [arXiv:2310.05164](#) | [DOI:10.1103/PhysRevLett.132.121901](#) | 2023-10-08

277. Muon identification using multivariate techniques in the CMS experiment in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

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[JINST 19 \(2024\) P02031](#) | [arXiv:2310.03844](#) | [DOI:10.1088/1748-0221/19/02/P02031](#) | 2023-10-05

278. Search for stealth supersymmetry in final states with two photons, jets, and low missing transverse momentum in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112009](#) | [arXiv:2310.03154](#) | [DOI:10.1103/PhysRevD.109.112009](#) | 2023-10-04

279. Study of azimuthal anisotropy of $\phi(1S)$ mesons in pPb collisions at $\sqrt{s_{NN}} = 8.16\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 850 \(2024\) 138518](#) | [arXiv:2310.03233](#) | [DOI:10.1016/j.physletb.2024.138518](#) | 2023-10-04

280. Search for a new resonance decaying into two spin-0 bosons in a final state with two photons and two bottom quarks in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 05 \(2024\) 316](#) | [arXiv:2310.01643](#) | [DOI:10.1007/JHEP05\(2024\)316](#) | 2023-10-02

281. Search for supersymmetry in final states with disappearing tracks in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 072007](#) | [arXiv:2309.16823](#) | [DOI:10.1103/PhysRevD.109.072007](#) | 2023-09-28

282. Search for direct production of GeV-scale resonances decaying to a pair of muons in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 12 \(2023\) 070](#) | [arXiv:2309.16003](#) | [DOI:10.1007/JHEP12\(2023\)070](#) | 2023-09-27

283. Inclusive and differential cross section measurements of ttbb production in the lepton+jets channel at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 042](#) | [arXiv:2309.14442](#) | [DOI:10.1007/JHEP05\(2024\)042](#) | 2023-09-25

284. Measurement of the τ lepton polarization in Z boson decays in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 01 \(2024\) 101](#) | [arXiv:2309.12408](#) | [DOI:10.1007/JHEP01\(2024\)101](#) | 2023-09-21

285. Development of the CMS detector for the CERN LHC Run 3

Hayrapetyan, Aram et al.

[JINST 19 \(2024\) P05064](#) | [arXiv:2309.05466](#) | [DOI:10.1088/1748-0221/19/05/P05064](#) | 2023-09-11

286. Evidence for the Higgs Boson Decay to a Z Boson and a Photon at the LHC

Aad, Georges et al.

[Phys.Rev.Lett. 132 \(2024\) 021803](#) | [arXiv:2309.03501](#) | [DOI:10.1103/PhysRevLett.132.021803](#) | 2023-09-07

287. Luminosity determination using Z boson production at the CMS experiment

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 26](#) | [arXiv:2309.01008](#) | [DOI:10.1140/epjc/s10052-023-12268-2](#) | 2023-09-02

288. Search for a third-generation leptoquark coupled to a τ lepton and a b quark through single, pair, and nonresonant production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 311](#) | [arXiv:2308.07826](#) | [DOI:10.1007/JHEP05\(2024\)311](#) | 2023-08-15

289. Search for Scalar Leptoquarks Produced via τ -Lepton-Quark Scattering in pp Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 061801](#) | [arXiv:2308.06143](#) | [DOI:10.1103/PhysRevLett.132.061801](#) | 2023-08-11

290. Measurement of the production cross section for a W boson in association with a charm quark in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 84 \(2024\) 27](#) | [arXiv:2308.02285](#) | [DOI:10.1140/epjc/s10052-023-12258-4](#) | 2023-08-04

291. Measurement of the Higgs boson production via vector boson fusion and its decay into bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 01 \(2024\) 173](#) | [arXiv:2308.01253](#) | [DOI:10.1007/JHEP01\(2024\)173](#) | 2023-08-02

292. Search for new physics in multijet events with at least one photon and large missing transverse momentum in proton-proton collisions at 13 TeV

Hayrapetyan, Aram et al.

[JHEP 10 \(2023\) 046](#) | [arXiv:2307.16216](#) | [DOI:10.1007/JHEP10\(2023\)046](#) | 2023-07-30

293. Search for physics beyond the standard model in top quark production with additional leptons in the context of effective field theory

Hayrapetyan, Aram et al.

[JHEP 12 \(2023\) 068](#) | [arXiv:2307.15761](#) | [DOI:10.1007/JHEP12\(2023\)068](#) | 2023-07-28

294. Multiplicity and transverse momentum dependence of charge-balance functions in pPb and PbPb collisions at LHC energies

Tumasyan, Armen et al.

[JHEP 08 \(2024\) 148](#) | [arXiv:2307.11185](#) | [DOI:10.1007/JHEP08\(2024\)148](#) | 2023-07-20

295. Study of charm hadronization with prompt Λ_c^+ baryons in proton-proton and lead-lead collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[JHEP 01 \(2024\) 128](#) | [arXiv:2307.11186](#) | [DOI:10.1007/JHEP01\(2024\)128](#) | 2023-07-20

296. Search for a high-mass dimuon resonance produced in association with b quark jets at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 10 \(2023\) 043](#) | [arXiv:2307.08708](#) | [DOI:10.1007/JHEP10\(2023\)043](#) | 2023-07-14

297. Search for Z' bosons decaying to pairs of heavy Majorana neutrinos in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2023\) 181](#) | [arXiv:2307.06959](#) | [DOI:10.1007/JHEP11\(2023\)181](#) | 2023-07-12

298. Evaluating Portable Parallelization Strategies for Heterogeneous Architectures in High Energy Physics

Atif, Mohammad et al.

[arXiv:2306.15869](#) | 2023-06-27

299. Detector R D needs for the next generation e^+e^- collider

Apresyan, A. et al.

[arXiv:2306.13567](#) | 2023-06-23

300. Two-particle Bose-Einstein correlations and their Lévy parameters in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

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[Phys.Rev.C 109 \(2024\) 024914](#) | [arXiv:2306.11574](#) | [DOI:10.1103/PhysRevC.109.024914](#) | 2023-06-20

301. Performance of the local reconstruction algorithms for the CMS hadron calorimeter with Run 2 data

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[JINST 18 \(2023\) P11017](#) | [arXiv:2306.10355](#) | [DOI:10.1088/1748-0221/18/11/P11017](#) | 2023-06-17

302. New Structures in the $J/\psi J/\psi$ Mass Spectrum in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 111901](#) | [arXiv:2306.07164](#) | [DOI:10.1103/PhysRevLett.132.111901](#) | 2023-06-12

303. Search for the lepton-flavor violating decay of the Higgs boson and additional Higgs bosons in the $e\mu$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 108 \(2023\) 072004](#) | [arXiv:2305.18106](#) | [DOI:10.1103/PhysRevD.108.072004](#) | 2023-05-29

304. Measurements of the azimuthal anisotropy of prompt and nonprompt charmonia in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[JHEP 10 \(2023\) 115](#) | [arXiv:2305.16928](#) | [DOI:10.1007/JHEP10\(2023\)115](#) | 2023-05-26

305. Observation of four top quark production in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 847 \(2023\) 138290](#) | [arXiv:2305.13439](#) | [DOI:10.1016/j.physletb.2023.138290](#) | 2023-05-22

306. Search for Inelastic Dark Matter in Events with Two Displaced Muons and Missing Transverse Momentum in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 041802](#) | [arXiv:2305.11649](#) | [DOI:10.1103/PhysRevLett.132.041802](#) | 2023-05-19

307. Search for resonances in events with photon and jet final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

JHEP 12 (2023) 189 | [arXiv:2305.07998](#) | [DOI:10.1007/JHEP12\(2023\)189](#) | 2023-05-13

308. Measurements of inclusive and differential cross sections for the Higgs boson production and decay to four-leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

JHEP 08 (2023) 040 | [arXiv:2305.07532](#) | [DOI:10.1007/JHEP08\(2023\)040](#) | 2023-05-12

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Balcas, Justas et al.

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310. Observation of the rare decay of the η meson to four muons

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311. Search for top squark pair production in a final state with at least one hadronically decaying tau lepton in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

JHEP 07 (2023) 110 | [arXiv:2304.07174](#) | [DOI:10.1007/JHEP07\(2023\)110](#) | 2023-04-14

312. Observation of the $\chi(3S)$ Meson and Suppression of χ States in Pb-Pb Collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

Phys.Rev.Lett. 133 (2024) 022302 | [arXiv:2303.17026](#) | [DOI:10.1103/PhysRevLett.133.022302](#) | 2023-03-29

313. Probing Small Bjorken-x Nuclear Gluonic Structure via Coherent J/ψ Photoproduction in Ultraperipheral Pb-Pb Collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

Phys.Rev.Lett. 131 (2023) 262301 | [arXiv:2303.16984](#) | [DOI:10.1103/PhysRevLett.131.262301](#) | 2023-03-29

314. First measurement of the top quark pair production cross section in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Tumasyan, Armen et al.

JHEP 08 (2023) 204 | [arXiv:2303.10680](#) | [DOI:10.1007/JHEP08\(2023\)204](#) | 2023-03-19

315. A search for new physics in central exclusive production using the missing mass technique with the CMS detector and the CMS-TOTEM precision proton spectrometer

Tumasyan, Armen et al.

Eur.Phys.J.C 83 (2023) 827 | [arXiv:2303.04596](#) | [DOI:10.1140/epjc/s10052-023-11687-5](#) | 2023-03-08

316. Evidence for Four-Top Quark Production in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

Phys.Lett.B 844 (2023) 138076 | [arXiv:2303.03864](#) | [DOI:10.1016/j.physletb.2023.138076](#) | 2023-03-07

317. A search for decays of the Higgs boson to invisible particles in events with a top-antitop quark pair or a vector boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

Eur.Phys.J.C 83 (2023) 933 | [arXiv:2303.01214](#) | [DOI:10.1140/epjc/s10052-023-11952-7](#) | 2023-03-02

318. Search for dark matter particles in W^+W^- events with transverse momentum imbalance in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

JHEP 03 (2024) 134 | [arXiv:2310.12229](#) | [DOI:10.1007/JHEP03\(2024\)134](#) | 2023-03

319. Search for a vector-like quark $T \rightarrow tH$ via the diphoton decay mode of the Higgs boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 09 \(2023\) 057](#) | [arXiv:2302.12802](#) | [DOI:10.1007/JHEP09\(2023\)057](#) | 2023-02-24

320. Measurement of the top quark mass using a profile likelihood approach with the lepton + jets final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 963](#) | [arXiv:2302.01967](#) | [DOI:10.1140/epjc/s10052-023-12050-4](#) | 2023-02-03

321. IRIS-HEP Strategic Plan for the Next Phase of Software Upgrades for HL-LHC Physics

Bockelman, Brian et al.

[arXiv:2302.01317](#) | 2023-02-02

322. Search for top squarks in the four-body decay mode with single lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2023\) 060](#) | [arXiv:2301.08096](#) | [DOI:10.1007/JHEP06\(2023\)060](#) | 2023-01-19

323. First Measurement of the Forward Rapidity Gap Distribution in pPb Collisions at $\sqrt{s_{NN}}=8.16$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 092004](#) | [arXiv:2301.07630](#) | [DOI:10.1103/PhysRevD.108.092004](#) | 2023-01-18

324. K_S^0 and $\Lambda(\Lambda^0)$ two-particle femtoscopic correlations in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 857 \(2024\) 138936](#) | [arXiv:2301.05290](#) | [DOI:10.1016/j.physletb.2024.138936](#) | 2023-01-12

325. Measurement of the electroweak production of $W\gamma$ in association with two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 032017](#) | [arXiv:2212.12592](#) | [DOI:10.1103/PhysRevD.108.032017](#) | 2022-12-23

326. Search for new physics in the τ lepton plus missing transverse momentum final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, A. et al.

[JHEP 09 \(2023\) 051](#) | [arXiv:2212.12604](#) | [DOI:10.1007/JHEP09\(2023\)051](#) | 2022-12-23

327. Measurement of the $B^0_S \rightarrow \mu^+ \mu^-$ decay properties and search for the $B^0 \rightarrow \mu^+ \mu^-$ decay in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137955](#) | [arXiv:2212.10311](#) | [DOI:10.1016/j.physletb.2023.137955](#) | 2022-12-20

328. Search for long-lived particles using out-of-time trackless jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 210](#) | [arXiv:2212.06695](#) | [DOI:10.1007/JHEP07\(2023\)210](#) | 2022-12-13

329. Measurement of the dependence of the hadron production fraction ratio f_s / f_u and f_d / f_u on B meson kinematic variables in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 121901](#) | [arXiv:2212.02309](#) | [DOI:10.1103/PhysRevLett.131.121901](#) | 2022-12-05

330. Measurements of azimuthal anisotropy of nonprompt D^0 mesons in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 850 \(2024\) 138389](#) | [arXiv:2212.01636](#) | [DOI:10.1016/j.physletb.2023.138389](#) | 2022-12-03

331. Search for high-mass exclusive $\gamma\gamma \rightarrow WW$ and $\gamma\gamma \rightarrow ZZ$ production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 229](#) | [arXiv:2211.16320](#) | [DOI:10.1007/JHEP07\(2023\)229](#) | 2022-11-29

332. Search for Higgs Boson and Observation of Z Boson through their Decay into a Charm Quark-Antiquark Pair in Boosted Topologies in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 041801](#) | [arXiv:2211.14181](#) | [DOI:10.1103/PhysRevLett.131.041801](#) | 2022-11-25

333. Search for supersymmetry in final states with a single electron or muon using angular correlations and heavy-object identification in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 09 \(2023\) 149](#) | [arXiv:2211.08476](#) | [DOI:10.1007/JHEP09\(2023\)149](#) | 2022-11-15

334. Measurement of the differential $t\bar{t}$ production cross section as a function of the jet mass and extraction of the top quark mass in hadronic decays of boosted top quarks

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 560](#) | [arXiv:2211.01456](#) | [DOI:10.1140/epjc/s10052-023-11587-8](#) | 2022-11-02

335. Azimuthal correlations in Z +jets events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 722](#) | [arXiv:2210.16139](#) | [DOI:10.1140/epjc/s10052-023-11833-z](#) | 2022-10-28

336. Measurements of jet multiplicity and jet transverse momentum in multijet events in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 742](#) | [arXiv:2210.13557](#) | [DOI:10.1140/epjc/s10052-023-11753-y](#) | 2022-10-24

337. Search for medium effects using jets from bottom quarks in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 844 \(2023\) 137849](#) | [arXiv:2210.08547](#) | [DOI:10.1016/j.physletb.2023.137849](#) | 2022-10-16

338. Azimuthal anisotropy of dijet events in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

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[JHEP 07 \(2023\) 139](#) | [arXiv:2210.08325](#) | [DOI:10.1007/JHEP07\(2023\)139](#) | 2022-10-15

339. The Future of High Energy Physics Software and Computing

Elvira, V. Daniel et al.

[arXiv:2210.05822](#) | [DOI:10.2172/1898754](#) | 2022-10-11

340. Proton reconstruction with the CMS-TOTEM Precision Proton Spectrometer

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[JINST 18 \(2023\) P09009](#) | [arXiv:2210.05854](#) | [DOI:10.1088/1748-0221/18/09/P09009](#) | 2022-10-11

341. Search for a heavy composite Majorana neutrino in events with dilepton signatures from proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 843 \(2023\) 137803](#) | [arXiv:2210.03082](#) | [DOI:10.1016/j.physletb.2023.137803](#) | 2022-10-06

342. Search for new heavy resonances decaying to WW , WZ , ZZ , WH , or ZH boson pairs in the all-jets final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 844 \(2023\) 137813](#) | [arXiv:2210.00043](#) | [DOI:10.1016/j.physletb.2023.137813](#) | 2022-09-30

343. Snowmass Computational Frontier: Topical Group Report on Experimental Algorithm Parallelization

Cerati, G. et al.

[arXiv:2209.07356](#) | 2022-09-15

344. Search for pair production of vector-like quarks in leptonic final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 07 \(2023\) 020](#) | [arXiv:2209.07327](#) | [DOI:10.1007/JHEP07\(2023\)020](#) | 2022-09-15

345. Search for exotic Higgs boson decays $H \rightarrow \mathcal{A}\mathcal{A} \rightarrow 4\gamma$ with events containing two merged diphotons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 101801](#) | [arXiv:2209.06197](#) | [DOI:10.1103/PhysRevLett.131.101801](#) | 2022-09-13

346. Search for new physics using effective field theory in 13 TeV pp collision events that contain a top quark pair and a boosted Z or Higgs boson

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 032008](#) | [arXiv:2208.12837](#) | [DOI:10.1103/PhysRevD.108.032008](#) | 2022-08-26

347. Measurement of the Higgs boson inclusive and differential fiducial production cross sections in the diphoton decay channel with pp collisions at $\sqrt{s} = 13$ TeV

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[JHEP 07 \(2023\) 091](#) | [arXiv:2208.12279](#) | [DOI:10.1007/JHEP07\(2023\)091](#) | 2022-08-25

348. Search for pair-produced vector-like leptons in final states with third-generation leptons and at least three b quark jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 846 \(2023\) 137713](#) | [arXiv:2208.09700](#) | [DOI:10.1016/j.physletb.2023.137713](#) | 2022-08-20

349. Measurement of the cross section of top quark-antiquark pair production in association with a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 07 \(2023\) 219](#) | [arXiv:2208.06485](#) | [DOI:10.1007/JHEP07\(2023\)219](#) | 2022-08-12

350. Measurement of the $t\bar{t}$ charge asymmetry in events with highly Lorentz-boosted top quarks in pp collisions at $\sqrt{s}=13$ TeV

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[Phys.Lett.B 846 \(2023\) 137703](#) | [arXiv:2208.02751](#) | [DOI:10.1016/j.physletb.2023.137703](#) | 2022-08-04

351. Searches for additional Higgs bosons and for vector leptoquarks in $\tau\tau$ final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 073](#) | [arXiv:2208.02717](#) | [DOI:10.1007/JHEP07\(2023\)073](#) | 2022-08-04

352. Search for CP violation in $t\bar{t}H$ and tH production in multilepton channels in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 092](#) | [arXiv:2208.02686](#) | [DOI:10.1007/JHEP07\(2023\)092](#) | 2022-08-04

353. Search for the exotic decay of the Higgs boson into two light pseudoscalars with four photons in the final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 148](#) | [arXiv:2208.01469](#) | [DOI:10.1007/JHEP07\(2023\)148](#) | 2022-08-02

354. Measurement of inclusive and differential cross sections for single top quark production in association with a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 046](#) | [arXiv:2208.00924](#) | [DOI:10.1007/JHEP07\(2023\)046](#) | 2022-08-01

355. Search for the Higgs boson decay to a pair of electrons in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Lett.B 846 \(2023\) 137783](#) | [arXiv:2208.00265](#) | [DOI:10.1016/j.physletb.2023.137783](#) | 2022-07-30

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Tumasyan, Armen et al.

[JHEP 07 \(2023\) 077](#) | [arXiv:2207.02270](#) | [DOI:10.1007/JHEP07\(2023\)077](#) | 2022-07-05

358. Search for direct pair production of supersymmetric partners of τ leptons in the final state with two hadronically decaying τ leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.D 108 \(2023\) 012011](#) | [arXiv:2207.02254](#) | [DOI:10.1103/PhysRevD.108.012011](#) | 2022-07-05

359. Search for a charged Higgs boson decaying into a heavy neutral Higgs boson and a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Nature 607 \(2022\) 60](#) | [arXiv:2207.00043](#) | [DOI:10.1038/s41586-022-04892-x](#) | 2022-06-30

361. Search for Higgs boson pairs decaying to WW^*WW^* , $WW^*\tau\tau$, and $\tau\tau\tau\tau$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 095](#) | [arXiv:2206.10268](#) | [DOI:10.1007/JHEP07\(2023\)095](#) | 2022-06-21

362. Search for nonresonant Higgs boson pair production in the four leptons plus two b jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2023\) 130](#) | [arXiv:2206.10657](#) | [DOI:10.1007/JHEP06\(2023\)130](#) | 2022-06-21

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Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 667](#) | [arXiv:2206.09466](#) | [DOI:10.1140/epjc/s10052-023-11632-6](#) | 2022-06-19

364. Search for nonresonant Higgs boson pair production in final state with two bottom quarks and two tau leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Lett.B 842 \(2023\) 137531](#) | [arXiv:2206.09401](#) | [DOI:10.1016/j.physletb.2022.137531](#) | 2022-06-19

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[Phys.Rev.Lett. 131 \(2023\) 011803](#) | [arXiv:2206.08956](#) | [DOI:10.1103/PhysRevLett.131.011803](#) | 2022-06-17

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[Phys.Lett.B 842 \(2023\) 137563](#) | [arXiv:2206.07110](#) | [DOI:10.1016/j.physletb.2022.137563](#) | 2022-06-14
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[Phys.Rev.Lett. 131 \(2023\) 151803](#) | [arXiv:2206.05192](#) | [DOI:10.1103/PhysRevLett.131.151803](#) | 2022-06-10
368. Search for Higgs boson decays into Z and J/ψ and for Higgs and Z boson decays into J/ψ or Y pairs in pp collisions at $\sqrt{s}=13$ TeV
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[Phys.Lett.B 842 \(2023\) 137534](#) | [arXiv:2206.03525](#) | [DOI:10.1016/j.physletb.2022.137534](#) | 2022-06-07
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[Phys.Rev.Lett. 131 \(2023\) 091803](#) | [arXiv:2206.02681](#) | [DOI:10.1103/PhysRevLett.131.091803](#) | 2022-06-06
370. Combination of inclusive top-quark pair production cross-section measurements using ATLAS and CMS data at $\sqrt{s} = 7$ and 8 TeV
Aad, G. et al.
[JHEP 07 \(2023\) 213](#) | [arXiv:2205.13830](#) | [DOI:10.1007/JHEP07\(2023\)213](#) | 2022-05-27
371. Search for electroweak production of charginos and neutralinos at $\sqrt{s}=13$ TeV in final states containing hadronic decays of WW, WZ, or WH and missing transverse momentum
Tumasyan, Armen et al.
[Phys.Lett.B 842 \(2023\) 137460](#) | [arXiv:2205.09597](#) | [DOI:10.1016/j.physletb.2022.137460](#) | 2022-05-19
372. Search for long-lived particles decaying to a pair of muons in proton-proton collisions at $\sqrt{s} = 13$ TeV
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[JHEP 05 \(2023\) 228](#) | [arXiv:2205.08582](#) | [DOI:10.1007/JHEP05\(2023\)228](#) | 2022-05-17
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[JHEP 07 \(2023\) 023](#) | [arXiv:2205.07434](#) | [DOI:10.1007/JHEP07\(2023\)023](#) | 2022-05-15
374. Search for heavy resonances and quantum black holes in $e\mu$, $e\tau$, and $\mu\tau$ final states in proton-proton collisions at $\sqrt{s} = 13$ TeV
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375. Search for Nonresonant Pair Production of Highly Energetic Higgs Bosons Decaying to Bottom Quarks
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[Phys.Rev.Lett. 131 \(2023\) 041803](#) | [arXiv:2205.06667](#) | [DOI:10.1103/PhysRevLett.131.041803](#) | 2022-05-13
376. Search for Higgs Boson Decay to a Charm Quark-Antiquark Pair in Proton-Proton Collisions at $\sqrt{s}=13$ TeV
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[Phys.Rev.D 108 \(2023\) 052004](#) | [arXiv:2205.02872](#) | [DOI:10.1103/PhysRevD.108.052004](#) | 2022-05-05

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[Eur.Phys.J.C 83 \(2023\) 587](#) | [arXiv:2205.02905](#) | [DOI:10.1140/epjc/s10052-023-11630-8](#) | 2022-05-05

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[Phys.Rev.Lett. 131 \(2023\) 051901](#) | [arXiv:2205.00045](#) | [DOI:10.1103/PhysRevLett.131.051901](#) | 2022-04-29

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[Eur.Phys.J.C 83 \(2023\) 571](#) | [arXiv:2204.13532](#) | [DOI:10.1140/epjc/s10052-023-11581-0](#) | 2022-04-28

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[Phys.Rev.D 108 \(2023\) 052002](#) | [arXiv:2204.12313](#) | [DOI:10.1103/PhysRevD.108.052002](#) | 2022-04-26

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[Phys.Lett.B 835 \(2022\) 137397](#) | [arXiv:2202.11807](#) | [DOI:10.1016/j.physletb.2022.137397](#) | 2022-02-23

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[Phys.Rev.Lett. 129 \(2022\) 081802](#) | [arXiv:2202.09617](#) | [DOI:10.1103/PhysRevLett.129.081802](#) | 2022-02-19

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[Nature Phys. 18 \(2022\) 1329](#) | [arXiv:2202.06923](#) | [DOI:10.1038/s41567-022-01682-0](#) | 2022-02-14

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[Eur.Phys.J.C 82 \(2022\) 499](#) | [arXiv:2201.09131](#) | [DOI:10.1140/epjc/s10052-022-10315-y](#) | 2022-01-22

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[Phys.Rev.Lett. 129 \(2022\) 021802](#) | [arXiv:2201.08476](#) | [DOI:10.1103/PhysRevLett.129.021802](#) | 2022-01-20

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[Phys.Rev.D 105 \(2022\) 072008](#) | [arXiv:2201.07861](#) | [DOI:10.1103/PhysRevD.105.072008](#) | 2022-01-19

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407. Search for long-lived heavy neutral leptons with displaced vertices in proton-proton collisions at $\sqrt{s} = 13$ TeV

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408. Search for higgsinos decaying to two Higgs bosons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 05 \(2022\) 014](#) | [arXiv:2201.04206](#) | [DOI:10.1007/JHEP05\(2022\)014](#) | 2022-01-11

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[Phys.Rev.Lett. 129 \(2022\) 022001](#) | [arXiv:2112.12236](#) | [DOI:10.1103/PhysRevLett.129.022001](#) | 2021-12-22

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[JHEP 06 \(2022\) 156](#) | [arXiv:2112.11125](#) | [DOI:10.1007/JHEP06\(2022\)156](#) | 2021-12-21

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417. Search for flavor-changing neutral current interactions of the top quark and the Higgs boson decaying to a bottom quark-antiquark pair at $\sqrt{s} = 13$ TeV

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[JHEP 02 \(2022\) 169](#) | [arXiv:2112.09734](#) | [DOI:10.1007/JHEP02\(2022\)169](#) | 2021-12-17

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419. Evidence for WW/WZ vector boson scattering in the decay channel $\ell\nu qq$ produced in association with two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Lett.B 834 \(2022\) 137438](#) | [arXiv:2112.05259](#) | [DOI:10.1016/j.physletb.2022.137438](#) | 2021-12-09

420. Search for a right-handed W boson and a heavy neutrino in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 04 \(2022\) 047](#) | [arXiv:2112.03949](#) | [DOI:10.1007/JHEP04\(2022\)047](#) | 2021-12-07

421. Search for heavy resonances decaying to a pair of Lorentz-boosted Higgs bosons in final states with leptons and a bottom quark pair at $\sqrt{s}=13$ TeV

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[JHEP 05 \(2022\) 005](#) | [arXiv:2112.03161](#) | [DOI:10.1007/JHEP05\(2022\)005](#) | 2021-12-06

422. Measurements of the associated production of a W boson and a charm quark in proton-proton collisions at $\sqrt{s}=8$ TeV

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[Eur.Phys.J.C 82 \(2022\) 1094](#) | [arXiv:2112.00895](#) | [DOI:10.1140/epjc/s10052-022-10897-7](#) | 2021-12-01

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[Phys.Rev.D 105 \(2022\) 052003](#) | [arXiv:2111.13948](#) | [DOI:10.1103/PhysRevD.105.052003](#) | 2021-11-27

424. Search for heavy resonances decaying to ZZ or ZW and axion-like particles mediating nonresonant ZZ or ZH production at $\sqrt{s} = 13$ TeV

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[JHEP 04 \(2022\) 087](#) | [arXiv:2111.13669](#) | [DOI:10.1007/JHEP04\(2022\)087](#) | 2021-11-26

425. Search for a heavy resonance decaying into a top quark and a W boson in the lepton+jets final state at $\sqrt{s} = 13$ TeV

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[JHEP 02 \(2022\) 142](#) | [arXiv:2111.10431](#) | [DOI:10.1007/JHEP02\(2022\)142](#) | 2021-11-19

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428. Search for supersymmetry in final states with two or three soft leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 04 \(2022\) 091](#) | [arXiv:2111.06296](#) | [DOI:10.1007/JHEP04\(2022\)091](#) | 2021-11-11

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[Nature Phys. 19 \(2023\) 338](#) | [arXiv:2111.05370](#) | [DOI:10.1038/s41567-022-01838-y](#) | 2021-11-09

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[JHEP 03 \(2022\) 189](#) | [arXiv:2111.04605](#) | [DOI:10.1007/JHEP03\(2022\)189](#) | 2021-11-08

431. Inclusive and differential cross section measurements of single top quark production in association with a Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 02 \(2022\) 107](#) | [arXiv:2111.02860](#) | [DOI:10.1007/JHEP02\(2022\)107](#) | 2021-11-04

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[JINST 17 \(2022\) P03014](#) | [arXiv:2111.03027](#) | [DOI:10.1088/1748-0221/17/03/P03014](#) | 2021-11-04

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[Phys.Rev.Lett. 129 \(2022\) 032001](#) | [arXiv:2111.02219](#) | [DOI:10.1103/PhysRevLett.129.032001](#) | 2021-11-03

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[Eur.Phys.J.C 82 \(2022\) 290](#) | [arXiv:2111.01299](#) | [DOI:10.1140/epjc/s10052-022-10127-0](#) | 2021-11-01

435. Search for long-lived particles produced in association with a Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 03 \(2022\) 160](#) | [arXiv:2110.13218](#) | [DOI:10.1007/JHEP03\(2022\)160](#) | 2021-10-25

436. Measurement of the inclusive and differential WZ production cross sections, polarization angles, and triple gauge couplings in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2022\) 032](#) | [arXiv:2110.11231](#) | [DOI:10.1007/JHEP07\(2022\)032](#) | 2021-10-21

437. First Search for Exclusive Diphoton Production at High Mass with Tagged Protons in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 129 \(2022\) 011801](#) | [arXiv:2110.05916](#) | [DOI:10.1103/PhysRevLett.129.011801](#) | 2021-10-12

438. Analysis of the CP structure of the Yukawa coupling between the Higgs boson and τ leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2022\) 012](#) | [arXiv:2110.04836](#) | [DOI:10.1007/JHEP06\(2022\)012](#) | 2021-10-10

439. Search for long-lived particles decaying to leptons with large impact parameter in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 153](#) | [arXiv:2110.04809](#) | [DOI:10.1140/epjc/s10052-022-10027-3](#) | 2021-10-10

440. Measurement of double-parton scattering in inclusive production of four jets with low transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 01 \(2022\) 177](#) | [arXiv:2109.13822](#) | [DOI:10.1007/JHEP01\(2022\)177](#) | 2021-09-28

441. Search for heavy resonances decaying to $Z(\nu\bar{\nu})V(q\bar{q}')$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 106 \(2022\) 012004](#) | [arXiv:2109.08268](#) | [DOI:10.1103/PhysRevD.106.012004](#) | 2021-09-17

442. Search for heavy resonances decaying to WW, WZ, or WH boson pairs in the lepton plus merged jet final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 032008](#) | [arXiv:2109.06055](#) | [DOI:10.1103/PhysRevD.105.032008](#) | 2021-09-13

443. Study of quark and gluon jet substructure in Z+jet and dijet events from pp collisions

Tumasyan, Armen et al.

[JHEP 01 \(2022\) 188](#) | [arXiv:2109.03340](#) | [DOI:10.1007/JHEP01\(2022\)188](#) | 2021-09-07

444. Observation of B^0_s mesons and measurement of the B^0_s/B^+ yield ratio in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 829 \(2022\) 137062](#) | [arXiv:2109.01908](#) | [DOI:10.1016/j.physletb.2022.137062](#) | 2021-09-04

445. Observation of tW production in the single-lepton channel in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 111](#) | [arXiv:2109.01706](#) | [DOI:10.1007/JHEP11\(2021\)111](#) | 2021-09-03

446. Measurement of the top quark mass using events with a single reconstructed top quark in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2021\) 161](#) | [arXiv:2108.10407](#) | [DOI:10.1007/JHEP12\(2021\)161](#) | 2021-08-23

447. Measurement of differential $t\bar{t}$ production cross sections in the full kinematic range using lepton+jets events from proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 104 \(2021\) 092013](#) | [arXiv:2108.02803](#) | [DOI:10.1103/PhysRevD.104.092013](#) | 2021-08-05

448. Probing effective field theory operators in the associated production of top quarks with a Z boson in multilepton final states at $\sqrt{s} = 13$ TeV

Lee, Kyeongpil et al.

[JHEP 12 \(2021\) 083](#) | [arXiv:2107.13896](#) | [DOI:10.1007/JHEP12\(2021\)083](#) | 2021-07-29

449. Search for new particles in events with energetic jets and large missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 153](#) | [arXiv:2107.13021](#) | [DOI:10.1007/JHEP11\(2021\)153](#) | 2021-07-27

450. Search for chargino-neutralino production in events with Higgs and W bosons using 137 fb^{-1} of proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 10 \(2021\) 045](#) | [arXiv:2107.12553](#) | [DOI:10.1007/JHEP10\(2021\)045](#) | 2021-07-27

451. Measurement of the inclusive and differential Higgs boson production cross sections in the decay mode to a pair of τ leptons in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 128 \(2022\) 081805](#) | [arXiv:2107.11486](#) | [DOI:10.1103/PhysRevLett.128.081805](#) | 2021-07-23

452. Combined searches for the production of supersymmetric top quark partners in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 81 \(2021\) 970](#) | [arXiv:2107.10892](#) | [DOI:10.1140/epjc/s10052-021-09721-5](#) | 2021-07-22

453. Search for Long-Lived Particles Decaying in the CMS End Cap Muon Detectors in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 127 \(2021\) 261804](#) | [arXiv:2107.04838](#) | [DOI:10.1103/PhysRevLett.127.261804](#) | 2021-07-10

454. Measurement of prompt open-charm production cross sections in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 225](#) | [arXiv:2107.01476](#) | [DOI:10.1007/JHEP11\(2021\)225](#) | 2021-07-03

455. Measurement of the inclusive and differential $t\bar{t}\gamma$ cross sections in the single-lepton channel and EFT interpretation at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2021\) 180](#) | [arXiv:2107.01508](#) | [DOI:10.1007/JHEP12\(2021\)180](#) | 2021-07-03

456. Measurements of the electroweak diboson production cross sections in proton-proton collisions at $\sqrt{s} = 5.02$ TeV using leptonic decays

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[Phys.Rev.Lett. 127 \(2021\) 191801](#) | [arXiv:2107.01137](#) | [DOI:10.1103/PhysRevLett.127.191801](#) | 2021-07-02

457. Search for electroweak production of charginos and neutralinos in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 147](#) | [arXiv:2106.14246](#) | [DOI:10.1007/JHEP04\(2022\)147](#) | 2021-06-27

458. Fragmentation of jets containing a prompt J/ψ meson in PbPb and pp collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 825 \(2022\) 136842](#) | [arXiv:2106.13235](#) | [DOI:10.1016/j.physletb.2021.136842](#) | 2021-06-23

459. Measurement of the electroweak production of $Z\gamma$ and two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV and constraints on anomalous quartic gauge couplings

Tumasyan, Armen et al.

[Phys.Rev.D 104 \(2021\) 072001](#) | [arXiv:2106.11082](#) | [DOI:10.1103/PhysRevD.104.072001](#) | 2021-06-21

460. Search for $W\gamma$ resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV using hadronic decays of Lorentz-boosted W bosons

Tumasyan, Armen et al.

[Phys.Lett.B 826 \(2022\) 136888](#) | [arXiv:2106.10509](#) | [DOI:10.1016/j.physletb.2022.136888](#) | 2021-06-19

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Tumasyan, Armen et al.

[JHEP 11 \(2021\) 057](#) | [arXiv:2106.10361](#) | [DOI:10.1007/JHEP11\(2021\)057](#) | 2021-06-18

462. Study of Z boson plus jets events using variables sensitive to double-parton scattering in pp collisions at 13 TeV

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 176](#) | [arXiv:2105.14511](#) | [DOI:10.1007/JHEP10\(2021\)176](#) | 2021-05-30

463. Measurements of the $pp \rightarrow W^\pm \gamma \gamma$ and $pp \rightarrow Z \gamma \gamma$ cross sections at $\sqrt{s} = 13$ TeV and limits on anomalous quartic gauge couplings

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 174](#) | [arXiv:2105.12780](#) | [DOI:10.1007/JHEP10\(2021\)174](#) | 2021-05-26

464. Search for strongly interacting massive particles generating trackless jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Eur.Phys.J.C 82 \(2022\) 213](#) | [arXiv:2105.09178](#) | [DOI:10.1140/epjc/s10052-022-10095-5](#) | 2021-05-19

465. Search for lepton-flavor violating decays of the Higgs boson in the $\mu\tau$ and $e\tau$ final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 032013](#) | [arXiv:2105.03007](#) | [DOI:10.1103/PhysRevD.104.032013](#) | 2021-05-06

466. Search for long-lived particles decaying to jets with displaced vertices in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A.M. et al.

[Phys.Rev.D 104 \(2021\) 052011](#) | [arXiv:2104.13474](#) | [DOI:10.1103/PhysRevD.104.052011](#) | 2021-04-27

467. Search for a heavy resonance decaying to a top quark and a W boson at $\sqrt{s} = 13$ TeV in the fully hadronic final state

Sirunyan, Albert M et al.

[JHEP 12 \(2021\) 106](#) | [arXiv:2104.12853](#) | [DOI:10.1007/JHEP12\(2021\)106](#) | 2021-04-26

468. Constraints on anomalous Higgs boson couplings to vector bosons and fermions in its production and decay using the four-lepton final state

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 052004](#) | [arXiv:2104.12152](#) | [DOI:10.1103/PhysRevD.104.052004](#) | 2021-04-25

469. Search for W' bosons decaying to a top and a bottom quark at $\sqrt{s} = 13$ TeV in the hadronic final state

Sirunyan, Albert M et al.

[Phys.Lett.B 820 \(2021\) 136535](#) | [arXiv:2104.04831](#) | [DOI:10.1016/j.physletb.2021.136535](#) | 2021-04-10

470. Search for charged Higgs bosons produced in vector boson fusion processes and decaying into vector boson pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 723](#) | [arXiv:2104.04762](#) | [DOI:10.1140/epjc/s10052-021-09472-3](#) | 2021-04-10

471. Precision luminosity measurement in proton-proton collisions at $\sqrt{s} = 13$ TeV in 2015 and 2016 at CMS

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 800](#) | [arXiv:2104.01927](#) | [DOI:10.1140/epjc/s10052-021-09538-2](#) | 2021-04-05

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Sirunyan, Albert M et al.

[Phys.Rev.Lett. 127 \(2021\) 102002](#) | [arXiv:2103.14089](#) | [DOI:10.1103/PhysRevLett.127.102002](#) | 2021-03-25

473. Measurements of Higgs boson production cross sections and couplings in the diphoton decay channel at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2021\) 027](#) | [arXiv:2103.06956](#) | [DOI:10.1007/JHEP07\(2021\)027](#) | 2021-03-11

474. Measurements of production cross sections of the Higgs boson in the four-lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 488](#) | [arXiv:2103.04956](#) | [DOI:10.1140/epjc/s10052-021-09200-x](#) | 2021-03-08

475. Using Z Boson Events to Study Parton-Medium Interactions in Pb-Pb Collisions

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 128 \(2022\) 122301](#) | [arXiv:2103.04377](#) | [DOI:10.1103/PhysRevLett.128.122301](#) | 2021-03-07

476. Search for resonant and nonresonant new phenomena in high-mass dilepton final states at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2021\) 208](#) | [arXiv:2103.02708](#) | [DOI:10.1007/JHEP07\(2021\)208](#) | 2021-03-03

477. Search for top squark production in fully-hadronic final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 052001](#) | [arXiv:2103.01290](#) | [DOI:10.1103/PhysRevD.104.052001](#) | 2021-03-01

478. Study of Drell-Yan dimuon production in proton-lead collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 182](#) | [arXiv:2102.13648](#) | [DOI:10.1007/JHEP05\(2021\)182](#) | 2021-02-26

479. Evidence for X(3872) in Pb-Pb Collisions and Studies of its Prompt Production at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M. et al.

[Phys.Rev.Lett. 128 \(2022\) 032001](#) | [arXiv:2102.13048](#) | [DOI:10.1103/PhysRevLett.128.032001](#) | 2021-02-25

480. First measurement of large area jet transverse momentum spectra in heavy-ion collisions

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 284](#) | [arXiv:2102.13080](#) | [DOI:10.1007/JHEP05\(2021\)284](#) | 2021-02-25

481. Measurements of angular distance and momentum ratio distributions in three-jet and Z + two-jet final states in pp collisions

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[Eur.Phys.J.C 81 \(2021\) 852](#) | [arXiv:2102.08816](#) | [DOI:10.1140/epjc/s10052-021-09570-2](#) | 2021-02-17

482. Search for a heavy vector resonance decaying to a Z^0 boson and a Higgs boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 688](#) | [arXiv:2102.08198](#) | [DOI:10.1140/epjc/s10052-021-09348-6](#) | 2021-02-16

483. Hard color-singlet exchange in dijet events in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.D 104 \(2021\) 032009](#) | [arXiv:2102.06945](#) | [DOI:10.1103/PhysRevD.104.032009](#) | 2021-02-13

484. Search for top squarks in final states with two top quarks and several light-flavor jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.D 104 \(2021\) 032006](#) | [arXiv:2102.06976](#) | [DOI:10.1103/PhysRevD.104.032006](#) | 2021-02-13

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[JINST 16 \(2021\) P07001](#) | [arXiv:2102.04790](#) | [DOI:10.1088/1748-0221/16/07/P07001](#) | 2021-02-09

486. Observation of a New Excited Beauty Strange Baryon Decaying to $\Xi^-_b \pi^+ \pi^-$

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[Phys.Rev.Lett. 126 \(2021\) 252003](#) | [arXiv:2102.04524](#) | [DOI:10.1103/PhysRevLett.126.252003](#) | 2021-02-08

487. Measurements of the differential cross sections of the production of Z + jets and γ + jets and of Z boson emission collinear with a jet in pp collisions at $\sqrt{s} = 13$ TeV

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488. Measurement of the $W\gamma$ Production Cross Section in Proton-Proton Collisions at $\sqrt{s}=13$ TeV and Constraints on Effective Field Theory Coefficients

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489. In-medium modification of dijets in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 116](#) | [arXiv:2101.04720](#) | [DOI:10.1007/JHEP05\(2021\)116](#) | 2021-01-12

490. First measurement of the cross section for top quark pair production with additional charm jets using dileptonic final states in pp collisions at $s=13\text{TeV}$

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[Phys.Lett.B 820 \(2021\) 136565](#) | [arXiv:2012.09225](#) | [DOI:10.1016/j.physletb.2021.136565](#) | 2020-12-16

491. Measurement of the Z boson differential production cross section using its invisible decay mode ($Z\nu\bar{\nu}$) in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

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492. Search for supersymmetry in final states with two oppositely charged same-flavor leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 04 \(2021\) 123](#) | [arXiv:2012.08600](#) | [DOI:10.1007/JHEP04\(2021\)123](#) | 2020-12-15

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495. Search for singly and pair-produced leptoquarks coupling to third-generation fermions in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

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Sirunyan, Albert M et al.
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[Phys.Rev.D 104 \(2021\) 012015](#) | [arXiv:2012.01581](#) | [DOI:10.1103/PhysRevD.104.012015](#) | 2020-12-02
498. Search for nonresonant Higgs boson pair production in final states with two bottom quarks and two photons in proton-proton collisions at $\sqrt{s} = 13$ TeV
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499. Search for the rare decay of the W boson into a pion and a photon in proton-proton collisions at $\sqrt{s}=13$ TeV
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[Phys.Lett.B 819 \(2021\) 136409](#) | [arXiv:2011.06028](#) | [DOI:10.1016/j.physletb.2021.136409](#) | 2020-11-11
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[Phys.Rev.Lett. 127 \(2021\) 122001](#) | [arXiv:2011.05239](#) | [DOI:10.1103/PhysRevLett.127.122001](#) | 2020-11-10
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Sirunyan, Albert M et al.
[Eur.Phys.J.C 81 \(2021\) 312](#) | [arXiv:2011.03422](#) | [DOI:10.1140/epjc/s10052-021-08949-5](#) | 2020-11-06
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Khachatryan, Vardan et al.
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Sirunyan, Albert M et al.
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506. Search for dark photons in Higgs boson production via vector boson fusion in proton-proton collisions at $\sqrt{s} = 13$ TeV
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[Eur.Phys.J.C 79 \(2019\) 269](#) | [arXiv:1811.10021](#) | [DOI:10.1140/epjc/s10052-019-6752-1](#) | 2018-11-25

685. Combination of searches for Higgs boson pair production in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Rev.Lett. 122 \(2019\) 121803](#) | [arXiv:1811.09689](#) | [DOI:10.1103/PhysRevLett.122.121803](#) | 2018-11-23

686. Search for associated production of a Higgs boson and a single top quark in proton-proton collisions at $\sqrt{s} = 13$ TeV
 Sirunyan, Albert M et al.
[Phys.Rev.D 99 \(2019\) 092005](#) | [arXiv:1811.09696](#) | [DOI:10.1103/PhysRevD.99.092005](#) | 2018-11-23
687. Search for resonant production of second-generation sleptons with same-sign dimuon events in proton-proton collisions at $\sqrt{s} = 13$ TeV
 Sirunyan, Albert M et al.
[Eur.Phys.J.C 79 \(2019\) 305](#) | [arXiv:1811.09760](#) | [DOI:10.1140/epjc/s10052-019-6800-x](#) | 2018-11-23
688. Search for a standard model-like Higgs boson in the mass range between 70 and 110 GeV in the diphoton final state in proton-proton collisions at $\sqrt{s}= 8$ and 13 TeV
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[Phys.Lett.B 793 \(2019\) 320](#) | [arXiv:1811.08459](#) | [DOI:10.1016/j.physletb.2019.03.064](#) | 2018-11-20
689. Search for long-lived particles decaying into displaced jets in proton-proton collisions at $\sqrt{s}= 13$ TeV
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[Phys.Rev.D 99 \(2019\) 032011](#) | [arXiv:1811.07991](#) | [DOI:10.1103/PhysRevD.99.032011](#) | 2018-11-19
690. Search for a W' boson decaying to a vector-like quark and a top or bottom quark in the all-jets final state
 Sirunyan, Albert M et al.
[JHEP 03 \(2019\) 127](#) | [arXiv:1811.07010](#) | [DOI:10.1007/JHEP03\(2019\)127](#) | 2018-11-16
691. Search for dark matter produced in association with a Higgs boson decaying to a pair of bottom quarks in proton-proton collisions at $\sqrt{s}=13$ TeV
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[Eur.Phys.J.C 79 \(2019\) 280](#) | [arXiv:1811.06562](#) | [DOI:10.1140/epjc/s10052-019-6730-7](#) | 2018-11-15
692. Measurements of $t\bar{t}$ differential cross sections in proton-proton collisions at $\sqrt{s}= 13$ TeV using events containing two leptons
 Sirunyan, Albert M et al.
[JHEP 02 \(2019\) 149](#) | [arXiv:1811.06625](#) | [DOI:10.1007/JHEP02\(2019\)149](#) | 2018-11-15
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 Sirunyan, Albert M et al.
[JHEP 04 \(2019\) 015](#) | [arXiv:1811.03052](#) | [DOI:10.1007/JHEP04\(2019\)015](#) | 2018-11-07
694. Search for pair production of first-generation scalar leptoquarks at $\sqrt{s} = 13$ TeV
 Sirunyan, Albert M et al.
[Phys.Rev.D 99 \(2019\) 052002](#) | [arXiv:1811.01197](#) | [DOI:10.1103/PhysRevD.99.052002](#) | 2018-11-03
695. Search for heavy neutrinos and third-generation leptoquarks in hadronic states of two τ leptons and two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV
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[JHEP 12 \(2018\) 117](#) | [arXiv:1811.00588](#) | [DOI:10.1007/JHEP12\(2018\)117](#) | 2018-11-01
697. Search for low-mass resonances decaying into bottom quark-antiquark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV
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[Phys.Rev.D 99 \(2019\) 012005](#) | [arXiv:1810.11822](#) | [DOI:10.1103/PhysRevD.99.012005](#) | 2018-10-28

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[Phys.Rev.Lett. 123 \(2019\) 022001](#) | [arXiv:1810.11102](#) | [DOI:10.1103/PhysRevLett.123.022001](#) | 2018-10-25

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[Phys.Rev.D 99 \(2019\) 012010](#) | [arXiv:1810.10092](#) | [DOI:10.1103/PhysRevD.99.012010](#) | 2018-10-23

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[Eur.Phys.J.C 79 \(2019\) 421](#) | [arXiv:1809.10733](#) | [DOI:10.1140/epjc/s10052-019-6909-y](#) | 2018-09-27

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[Phys.Lett.B 793 \(2019\) 520](#) | [arXiv:1809.05937](#) | [DOI:10.1016/j.physletb.2019.04.025](#) | 2018-09-16

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[Phys.Rev.Lett. 121 \(2018\) 241802](#) | [arXiv:1809.05558](#) | [DOI:10.1103/PhysRevLett.121.241802](#) | 2018-09-14

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[Phys.Lett.B 792 \(2019\) 345](#) | [arXiv:1808.03684](#) | [DOI:10.1016/j.physletb.2019.01.072](#) | 2018-08-10

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[Phys.Rev.D 98 \(2018\) 112014](#) | [arXiv:1808.03124](#) | [DOI:10.1103/PhysRevD.98.112014](#) | 2018-08-09

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[Phys.Rev.Lett. 121 \(2018\) 221802](#) | [arXiv:1808.02913](#) | [DOI:10.1103/PhysRevLett.121.221802](#) | 2018-08-08

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[Phys.Rev.Lett. 122 \(2019\) 081804](#) | [arXiv:1808.01257](#) | [DOI:10.1103/PhysRevLett.122.081804](#) | 2018-08-03

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[Phys.Lett.B 792 \(2019\) 107](#) | [arXiv:1807.11421](#) | [DOI:10.1016/j.physletb.2019.01.069](#) | 2018-07-30

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[Phys.Rev.Lett. 122 \(2019\) 011803](#) | [arXiv:1807.06522](#) | [DOI:10.1103/PhysRevLett.122.011803](#) | 2018-07-17

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[JINST 13 \(2018\) P10034](#) | [arXiv:1807.03289](#) | [DOI:10.1088/1748-0221/13/10/P10034](#) | 2018-07-09

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[Eur.Phys.J.C 79 \(2019\) 123](#) | [arXiv:1807.02810](#) | [DOI:10.1140/epjc/s10052-019-6620-z](#) | 2018-07-08

742. Search for heavy resonances decaying into a vector boson and a Higgs boson in final states with charged leptons, neutrinos and b quarks at $\sqrt{s} = 13$ TeV

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[JHEP 11 \(2018\) 172](#) | [arXiv:1807.02826](#) | [DOI:10.1007/JHEP11\(2018\)172](#) | 2018-07-08

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[Eur.Phys.J.C 79 \(2019\) 20](#) | [arXiv:1807.00782](#) | [DOI:10.1140/epjc/s10052-018-6482-9](#) | 2018-07-02

745. Measurement of charged particle spectra in minimum-bias events from proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Eur.Phys.J.C 78 \(2018\) 697](#) | [arXiv:1806.11245](#) | [DOI:10.1140/epjc/s10052-018-6144-y](#) | 2018-06-29

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Sirunyan, Albert M. et al.

[Phys.Lett.B 789 \(2019\) 19](#) | [arXiv:1806.11073](#) | [DOI:10.1016/j.physletb.2018.11.007](#) | 2018-06-28

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750. Search for supersymmetric partners of electrons and muons in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Phys.Lett.B 790 \(2019\) 140](#) | [arXiv:1806.05264](#) | [DOI:10.1016/j.physletb.2019.01.005](#) | 2018-06-13

751. Search for dark matter produced in association with a Higgs boson decaying to $\gamma\gamma$ or $\tau^+\tau^-$ at $\sqrt{s} = 13$ TeV

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[JHEP 09 \(2018\) 046](#) | [arXiv:1806.04771](#) | [DOI:10.1007/JHEP09\(2018\)046](#) | 2018-06-12

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[Phys.Rev.Lett. 121 \(2018\) 141801](#) | [arXiv:1806.04213](#) | [DOI:10.1103/PhysRevLett.121.141801](#) | 2018-06-11

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754. Search for resonant pair production of Higgs bosons decaying to bottom quark-antiquark pairs in proton-proton collisions at 13 TeV

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[Phys.Rev.Lett. 121 \(2018\) 141802](#) | [arXiv:1806.01058](#) | [DOI:10.1103/PhysRevLett.121.141802](#) | 2018-06-04

756. Search for narrow and broad dijet resonances in proton-proton collisions at $\sqrt{s}=13$ TeV and constraints on dark matter mediators and other new particles

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[JHEP 08 \(2018\) 130](#) | [arXiv:1806.00843](#) | [DOI:10.1007/JHEP08\(2018\)130](#) | 2018-06-03

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[Phys.Rev.D 98 \(2018\) 112011](#) | [arXiv:1806.00636](#) | [DOI:10.1103/PhysRevD.98.112011](#) | 2018-06-02

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[Phys.Lett.B 788 \(2019\) 7](#) | [arXiv:1806.00408](#) | [DOI:10.1016/j.physletb.2018.10.056](#) | 2018-06-01

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762. Search for an exotic decay of the Higgs boson to a pair of light pseudoscalars in the final state with two b quarks and two τ leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Lett.B 785 \(2018\) 462](#) | [arXiv:1805.10191](#) | [DOI:10.1016/j.physletb.2018.08.057](#) | 2018-05-25

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[Phys.Rev.D 98 \(2018\) 032005](#) | [arXiv:1805.10228](#) | [DOI:10.1103/PhysRevD.98.032005](#) | 2018-05-25

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[Phys.Lett.B 790 \(2019\) 270](#) | [arXiv:1805.09215](#) | [DOI:10.1016/j.physletb.2019.01.006](#) | 2018-05-23

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[JHEP 10 \(2018\) 117](#) | [arXiv:1805.07399](#) | [DOI:10.1007/JHEP10\(2018\)117](#) | 2018-05-18

766. Search for top squarks decaying via four-body or chargino-mediated modes in single-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[JHEP 09 \(2018\) 065](#) | [arXiv:1805.05784](#) | [DOI:10.1007/JHEP09\(2018\)065](#) | 2018-05-15

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Sirunyan, Albert M et al.

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[JHEP 11 \(2018\) 018](#) | [arXiv:1805.04865](#) | [DOI:10.1007/JHEP11\(2018\)018](#) | 2018-05-13

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Sirunyan, Albert M et al.

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[JHEP 08 \(2018\) 016](#) | [arXiv:1804.07321](#) | [DOI:10.1007/JHEP08\(2018\)016](#) | 2018-04-19

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Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 965](#) | [arXiv:1804.05252](#) | [DOI:10.1140/epjc/s10052-018-6373-0](#) | 2018-04-14

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[JINST 13 \(2018\) P06015](#) | [arXiv:1804.04528](#) | [DOI:10.1088/1748-0221/13/06/P06015](#) | 2018-04-12

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Sirunyan, Albert M et al.

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[JHEP 09 \(2018\) 101](#) | [arXiv:1803.10093](#) | [DOI:10.1007/JHEP09\(2018\)101](#) | 2018-03-27

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Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 112003](#) | [arXiv:1803.08856](#) | [DOI:10.1103/PhysRevD.97.112003](#) | 2018-03-23

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[Eur.Phys.J.C 78 \(2018\) 789](#) | [arXiv:1803.08030](#) | [DOI:10.1140/epjc/s10052-018-6242-x](#) | 2018-03-21

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[JHEP 08 \(2018\) 066](#) | [arXiv:1803.05485](#) | [DOI:10.1007/JHEP08\(2018\)066](#) | 2018-03-14

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[JHEP 07 \(2018\) 153](#) | [arXiv:1803.04496](#) | [DOI:10.1007/JHEP07\(2018\)153](#) | 2018-03-12

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Sirunyan, A.M. et al.

[JHEP 06 \(2018\) 002](#) | [arXiv:1803.03991](#) | [DOI:10.1007/JHEP06\(2018\)002](#) | 2018-03-11

794. Search for a heavy resonance decaying into a Z boson and a vector boson in the $\nu\nu qq$ final state

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[JHEP 07 \(2018\) 075](#) | [arXiv:1803.03838](#) | [DOI:10.1007/JHEP07\(2018\)075](#) | 2018-03-10

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[Eur.Phys.J.C 78 \(2018\) 707](#) | [arXiv:1803.02864](#) | [DOI:10.1140/epjc/s10052-018-6143-z](#) | 2018-03-07

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[Phys.Rev.D 97 \(2018\) 072010](#) | [arXiv:1802.04867](#) | [DOI:10.1103/PhysRevD.97.072010](#) | 2018-02-13

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[Phys.Rev.Lett. 120 \(2018\) 221801](#) | [arXiv:1802.02965](#) | [DOI:10.1103/PhysRevLett.120.221801](#) | 2018-02-08

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[JHEP 07 \(2018\) 161](#) | [arXiv:1802.02613](#) | [DOI:10.1007/JHEP07\(2018\)161](#) | 2018-02-07

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Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 025](#) | [arXiv:1802.02110](#) | [DOI:10.1007/JHEP05\(2018\)025](#) | 2018-02-06

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[JHEP 06 \(2018\) 031](#) | [arXiv:1802.01486](#) | [DOI:10.1007/JHEP06\(2018\)031](#) | 2018-02-05

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[JHEP 06 \(2018\) 027](#) | [arXiv:1801.08427](#) | [DOI:10.1007/JHEP06\(2018\)027](#) | 2018-01-25

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[Phys.Rev.Lett. 121 \(2018\) 242301](#) | [arXiv:1801.04895](#) | [DOI:10.1103/PhysRevLett.121.242301](#) | 2018-01-15

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[JHEP 03 \(2018\) 160](#) | [arXiv:1801.03957](#) | [DOI:10.1007/JHEP03\(2018\)160](#) | 2018-01-11

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[Eur.Phys.J.C 78 \(2018\) 708](#) | [arXiv:1801.03535](#) | [DOI:10.1140/epjc/s10052-018-6146-9](#) | 2018-01-10

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[Phys.Lett.B 783 \(2018\) 114](#) | [arXiv:1712.08920](#) | [DOI:10.1016/j.physletb.2018.06.028](#) | 2017-12-24

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[Eur.Phys.J.C 78 \(2018\) 509](#) | [arXiv:1712.08959](#) | [DOI:10.1140/epjc/s10052-018-5950-6](#) | 2017-12-24

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[Phys.Rev.Lett. 120 \(2018\) 241801](#) | [arXiv:1712.08501](#) | [DOI:10.1103/PhysRevLett.120.241801](#) | 2017-12-22

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[JINST 13 \(2018\) P05011](#) | [arXiv:1712.07158](#) | [DOI:10.1088/1748-0221/13/05/P05011](#) | 2017-12-19
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Sirunyan, A. M. et al.
[Phys.Rev.D 97 \(2018\) 092005](#) | [arXiv:1712.02345](#) | [DOI:10.1103/PhysRevD.97.092005](#) | 2017-12-06
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[Phys.Lett.B 778 \(2018\) 349](#) | [arXiv:1711.10949](#) | [DOI:10.1016/j.physletb.2018.01.049](#) | 2017-11-29
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[Phys.Lett.B 780 \(2018\) 118](#) | [arXiv:1711.08008](#) | [DOI:10.1016/j.physletb.2018.02.045](#) | 2017-11-21

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[Phys.Lett.B 789 \(2019\) 643](#) | [arXiv:1711.05594](#) | [DOI:10.1016/j.physletb.2018.11.063](#) | 2017-11-15

833. Search for excited quarks of light and heavy flavor in γ + jet final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Phys.Lett.B 781 \(2018\) 390](#) | [arXiv:1711.04652](#) | [DOI:10.1016/j.physletb.2018.04.007](#) | 2017-11-13

834. Measurement of the underlying event activity in inclusive Z boson production in proton-proton collisions at $\sqrt{s}=13$ TeV

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[JHEP 03 \(2018\) 003](#) | [arXiv:1711.04370](#) | [DOI:10.1007/JHEP03\(2018\)003](#) | 2017-11-12

836. Measurement of the inclusive tt cross section in pp collisions at $\sqrt{s}=5.02$ TeV using final states with at least one charged lepton

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[Eur.Phys.J.C 78 \(2018\) 287](#) | [arXiv:1711.02143](#) | [DOI:10.1140/epjc/s10052-018-5752-x](#) | 2017-11-06

839. Search for top squarks and dark matter particles in opposite-charge dilepton final states at $\sqrt{s}= 13$ TeV

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842. Measurement of quarkonium production cross sections in pp collisions at $\sqrt{s} = 13$ TeV

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[Phys.Lett.B 780 \(2018\) 251](#) | [arXiv:1710.11002](#) | [DOI:10.1016/j.physletb.2018.02.033](#) | 2017-10-30

843. Search for supersymmetry in proton-proton collisions at 13 TeV using identified top quarks

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[Phys.Rev.D 97 \(2018\) 012007](#) | [arXiv:1710.11188](#) | [DOI:10.1103/PhysRevD.97.012007](#) | 2017-10-30

844. Search for standard model production of four top quarks with same-sign and multilepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[Eur.Phys.J.C 78 \(2018\) 140](#) | [arXiv:1710.10614](#) | [DOI:10.1140/epjc/s10052-018-5607-5](#) | 2017-10-29

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[JHEP 01 \(2018\) 045](#) | [arXiv:1710.09355](#) | [DOI:10.1007/JHEP01\(2018\)045](#) | 2017-10-25

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[Eur.Phys.J.C 78 \(2018\) 457](#) | [arXiv:1710.08949](#) | [DOI:10.1140/epjc/s10052-018-5929-3](#) | 2017-10-24

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[JHEP 03 \(2018\) 172](#) | [arXiv:1710.07955](#) | [DOI:10.1007/JHEP03\(2018\)172](#) | 2017-10-22

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Sirunyan, Albert M et al.

[Phys.Rev.C 98 \(2018\) 044902](#) | [arXiv:1710.07864](#) | [DOI:10.1103/PhysRevC.98.044902](#) | 2017-10-21

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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

[Phys.Lett.B 752 \(2016\) 221](#) | [arXiv:1508.01437](#) | [DOI:10.1016/j.physletb.2015.11.042](#) | 2015-08-06

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[Phys.Lett.B 757 \(2016\) 6](#) | [arXiv:1508.01218](#) | [DOI:10.1016/j.physletb.2016.03.039](#) | 2015-08-05

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[Phys.Lett.B 751 \(2015\) 143](#) | [arXiv:1507.05915](#) | [DOI:10.1016/j.physletb.2015.10.004](#) | 2015-07-21

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[Phys.Lett.B 757 \(2016\) 154](#) | [arXiv:1507.03119](#) | [DOI:10.1016/j.physletb.2016.03.060](#) | 2015-07-11

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[Phys.Lett.B 753 \(2016\) 341](#) | [arXiv:1507.03031](#) | [DOI:10.1016/j.physletb.2015.12.039](#) | 2015-07-10

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[Phys.Lett.B 748 \(2015\) 221](#) | [arXiv:1504.04710](#) | [DOI:10.1016/j.physletb.2015.07.010](#) | 2015-04-18

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Khachatryan, Vardan et al.

[Phys.Lett.B 749 \(2015\) 187](#) | [arXiv:1504.03511](#) | [DOI:10.1016/j.physletb.2015.07.065](#) | 2015-04-14

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Khachatryan, Vardan et al.

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Khachatryan, Vardan et al.

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[Phys.Rev.D 92 \(2015\) 012003](#) | [arXiv:1503.08689](#) | [DOI:10.1103/PhysRevD.92.012003](#) | 2015-03-30

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[Phys.Lett.B 750 \(2015\) 565](#) | [arXiv:1503.05825](#) | [DOI:10.1016/j.physletb.2015.09.057](#) | 2015-03-19

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[Phys.Lett.B 749 \(2015\) 560](#) | [arXiv:1503.04114](#) | [DOI:10.1016/j.physletb.2015.08.047](#) | 2015-03-13

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[JHEP 06 \(2015\) 080](#) | [arXiv:1503.01952](#) | [DOI:10.1007/JHEP06\(2015\)080](#) | 2015-03-06

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Khachatryan, Vardan et al.
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[Eur.Phys.J.C 75 \(2015\) 151](#) | [arXiv:1501.05603](#) | [DOI:10.1140/epjc/s10052-015-3367-z](#) | 2015-01-22
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[Eur.Phys.J.C 75 \(2015\) 147](#) | [arXiv:1412.1115](#) | [DOI:10.1140/epjc/s10052-015-3364-2](#) | 2014-12-02

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Khachatryan, Vardan et al.

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Chatrchyan, Serguei et al.

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